🕸️ Service Mesh — Easy KCNA Notes

Since you're learning this as a **beginner**, first remember:

Service Mesh = a dedicated system that manages communication between services.

It is especially useful when you have many **microservices**.

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1. Why do we need a Service Mesh?

Imagine an application with many microservices:

User Service

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Order Service

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Payment Service

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Product Service

These services need to communicate with each other.

As the number of services increases, networking becomes difficult.

You may need:

- 🔐 Security
- 🚦 Traffic management
- 📊 Monitoring
- 🔁 Retries
- ❤️ Health checking
- ⚖️ Load balancing

A **Service Mesh** helps manage these communication-related tasks.

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2. Simple Real-Life Example 🚦

Imagine a city with thousands of cars.

You need:

🚦 Traffic lights

👮 Traffic police

🛣️ Roads

📊 Traffic monitoring

Similarly, in a microservices application:

Microservices

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Service Mesh

↓

Manages communication

Easy memory:

Service Mesh = Traffic management system for microservices.

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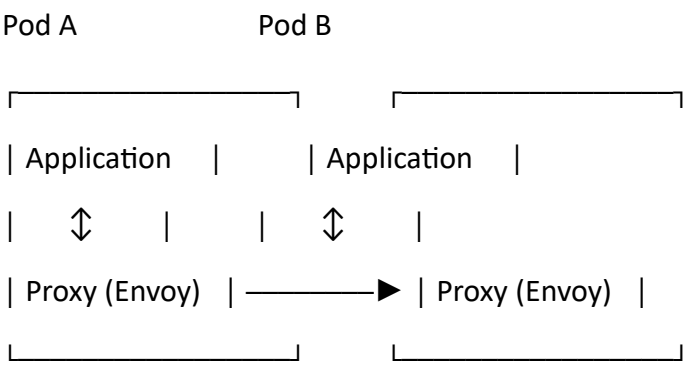
3. How does Service Mesh work?

A common service-mesh design uses a **proxy** alongside each application.

For example:



The application communicates through the proxy.

The proxy handles many networking concerns.

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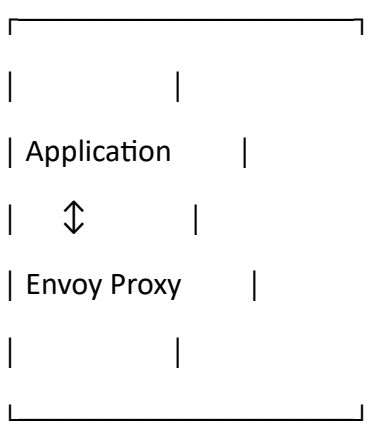
4. Envoy and Service Mesh ⭐

You just learned about **Envoy**.

This connects directly to Service Mesh.

Envoy can act as the **proxy** for a service.

Pod



So remember:

Envoy = Proxy

Service Mesh = System for managing service-to-service communication

Envoy is one technology commonly used to implement the proxy/data-plane part of a service mesh.

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5. What does a Service Mesh provide?

🔐 **Security**

Can help secure communication between services.

Example:

Service A 🔐 ———— 🔐 Service B

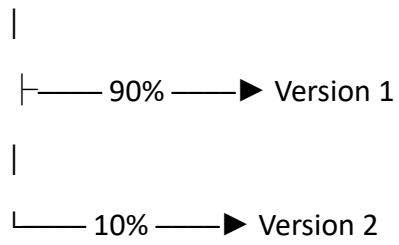
It can support encrypted service-to-service communication.

🚦 Traffic Management

It can control how traffic moves between services.

Example:

Service A

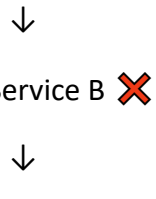


This can be useful for gradual releases.

🔄 Retries

If a request fails, the proxy can potentially retry it according to configured policies.

Request



⏱️ Timeouts

A service mesh can configure how long a request should wait.

Request

Wait

↓

Timeout ⏱️

This prevents requests from waiting forever.

📊 Observability

It can provide information about service communication:

- Requests
- Errors
- Latency
- Traffic

Example:

Service A → Service B

1,000 requests

20 errors

100 ms latency

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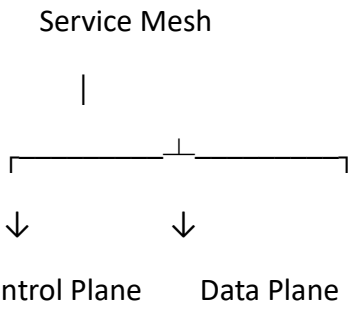
6. Control Plane and Data Plane ⭐⭐⭐

This is an important concept.

A service mesh is commonly divided into **two parts**:



🧠 Control Plane

The **control plane** manages and configures the proxies.

Think:

Control Plane = Brain 🧠

It tells proxies what rules/policies to follow.

🚦 **Data Plane**

The **data plane** handles the actual application traffic.

It consists of the proxies.

Think:

Data Plane = Traffic 🚦

Control Plane

  |

  | configuration

  ▼

Envoy Proxies

  |

  | actual traffic

  ▼

Microservices

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7. Service Mesh vs Kubernetes ⭐

Don't confuse them.

Kubernetes

Manages the **containers/workloads and cluster**.

Kubernetes

  ↓

Pods

Deployments

Services

Nodes

Service Mesh

Focuses mainly on **service-to-service communication**.

Service Mesh

↓

Traffic

Security

Observability

Retries

Timeouts

They can work **together**.

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8. Service Mesh vs Service

These are also different.

Kubernetes Service

Provides a stable way to reach a set of Pods.

Client

↓

Service

↓

Pods

Service Mesh

Manages communication **between services** and provides additional networking features.

Service A

↕

Service Mesh

↕

Service B

🧠 **KCNA Quick Revision**

Service Mesh

- 🕸️ Manages **service-to-service communication**
- Commonly used with **microservices**
- Helps with:
  - 🔒 Security
  - 🚦 Traffic management
  - 📊 Observability
  - 🔄 Retries
  - ⏱️ Timeouts
- Often uses **sidecar proxies**
- **Envoy** is a popular proxy used in service meshes.
- **Control Plane** → manages/configures proxies
- **Data Plane** → handles actual network traffic

⭐ **One-line definition**

A Service Mesh is a dedicated infrastructure layer that manages communication between microservices.

🧠 **Easy memory**

Kubernetes → manages your workloads

Service Mesh → manages communication between those workloads

Envoy → acts as the traffic proxy